An Efficient Detection and Classification of Acute Leukemia Using Transfer Learning and Orthogonal Softmax Layer-Based Model

Pradeep Kumar Das, Biswajeet Sahoo, and Sukadev Meher

Abstract—For the early diagnosis of hematological disorders like blood cancer, microscopic analysis of blood cells is very important. Traditional deep CNNs lead to overfitting when it receives small medical image datasets such as ALLIDB1, ALLIDB2, and ASH. This paper proposes a new and effective model for classifying and detecting Acute Lymphoblastic Leukemia (ALL) or Acute Myelogenous Leukemia (AML) that delivers excellent performance in small medical datasets. Here, we have proposed a novel Orthogonal SoftMax Layer (OSL)-based Acute Leukemia detection model that consists of ResNet 18-based deep feature extraction followed by efficient OSL-based classification. Here, OSL is integrated with the ResNet18 to improve the classification performance by making the weight vectors orthogonal to each other. Hence, it integrates ResNet benefits (residual learning and identity mapping) with the benefits of OSL-based classification (improvement of feature discrimination capability and computational efficiency). Furthermore, we have introduced extra dropout and ReLu layers in the architecture to achieve a faster network with enhanced performance. The performance verification is performed on standard ALLIDB1, ALLIDB2, and C_NMC_2019 datasets for efficient ALL detection and ASH dataset for effective AML detection. The experimental performance demonstrates the superiority of the proposed model over other compairing models.

Index Terms—Acute lymphoblastic leukemia, acute myelogenous leukemia, blood cancer, classification, orthogonal softMax layer (OSL), transfer learning

1 Introduction

Over the years, medical image processing research aims to suggest an automated and more accurate medical system that helps diagnose disease and treatment planning [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. The important factor to look after is how efficiently the system solves the problems in a particular medical application. Recently, transfer learning has emerged as a preferred scheme for effective small-sample classification because of its exemplary performance on small datasets [2], [3], [9], [11], [12], [13], [14], [15]. In this work, a novel OSL-based Acute Leukemia detection model is proposed that achieves excellent performance in small medical datasets.

Acute leukemia is a rapidly-growing blood cancer that emerges from the bone marrow and severely affects white blood cells (WBCs). Hence, leukemic patients face difficulties in their bodies' ability to fight against infections; thus affecting the immune system. It has the highest impact during the childhood period and old age [1], [2], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. As suggested by the French-American-British model, it is of two categories: ALL and AML. ALL spreads rapidly to damage lymphocytes and makes them lymphoblasts. It occurs when lymphoblasts gather in the bone marrow. It can also be fatal at times, affecting the liver, spleen, etc. [1], [2], [13], [14], [16], [17], [18]. It might have symptoms like fever, stomach pains, fatigue, and vomiting. AML affects the bone marrow and spreads rapidly to damage the non-lymphocytes (WBCs). It creates soft tissues in the bones. Its symptoms include fever, bone pain, bleeding, and bruising [19], [20], [21], [22]. It can also severely affect the brain and skin if not treated in the initial phase. Hence, early disease diagnosis with appropriate treatment facilities can improve health conditions and save lives.

Initially, researchers have developed some diagnosis systems which mainly employ traditional machine techniques. The system process usually starts with the preprocessing of the input image and further goes on to processes like segmentation, feature extraction, and finally have the classification stage [2], [5], [13], [14], [16], [17], [18], [19], [20], [21], [22]. However, to increase the efficiency of the system, Convolutional Neural Networks (CNNs) are employed nowadays. The traditional machine learning-based techniques usually use hand-crafted features to perform classification. On the other hand, modern CNN-based techniques extract deep features directly from the raw input images [2], [5], [9], [13], [14].

CNN is one of the most attractive developments in Computer Vision. It has applications in the real world like face recognition, image classification, visual search, object detection, and disease diagnosis [2], [4], [6], [9], [11], [12], [13], [14]. But the main issue which comes is the large training data size. A large amount of training data is required for

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proper training to yield efficient feature extraction/ classification [2], [9], [11], [12], [13], [14], [23], [24]. On the other hand, getting a labeled dataset is one big hurdle when it comes to medical image classifications. Dataset collection is the most costly, time-consuming task in the whole process. Many medical institutes also don't share their patients' data for obvious reasons like privacy. Hence, the dataset available is always limited in size. In traditional deep learning-based systems, if the dataset size is small, it tends to overfit and cause instability. So, a deep CNN (DCNN) with transfer learning has evolved [2], [9], [11], [12], [13], [14] to overcome such issues.

Due to the unavailability of large datasets, small-sample classification in the deep learning field is highly appreciated. However, the main challenge is to handle overfitting issues properly [11]. To solve this issue, several researchers have recommended various approaches: regularization [25], data augmentation [2], [26], [27], ensemble techniques [9], domain adaption [28], [29], and discriminative feature learning [11], [30], [31].

Currently, discriminative feature learning has grown as a preferred approach to boost deep learning-based classification performance. Though employment of virtual softmax technique or large margin loss can improve the performance, these techniques make the optimization more difficult (due to the addition of extra constraint on a loss function) or reduce the applicability of a type of data (due to assumptions on data) [11]. However, the objective of the proposed work is to get an appropriate subspace of CNNs, which can easily achieve high discriminative feature learning and a big decision margin. For this purpose, we propose a novel Orthogonal SoftMax Layer (OSL)-based Acute Leukemia detection model, which has been inspired by the OSLNet approach recommended by Li et al. [11]. Here, OSL replaces a fully connected classification layer (FCCL) to boost the classification performance.

1.1 Contribution

The contributions are given below.

1. A novel Orthogonal SoftMax Layer (OSL)-based Acute Leukemia detection model is proposed that delivers excellent performance in small medical datasets. It consists of ResNet 18-based deep feature extraction followed by efficient OSL-based classification.
2. Here, OSL is integrated with the ResNet18 to improve the classification performance by making the weight vectors orthogonal to each other in the classification layer during both the training and testing stages. It also retains the benefits of ResNet, i.e., residual learning and identity mapping.
3. In this model, OSL replaces a fully connected classification layer (FCCL) to make the network optimization easier by minimizing the co-adaption among parameters in the classification layer. Moreover, a model with OSL exhibits a low generalization error compared to an FCCL. It optimizes the number of connections (as a few connections are eliminated) and maximizes inter-class angles as weight-vectors of distinct classes are pairwise orthogonal.
4. Furthermore, extra dropout and ReLu layers are introduced in the proposed architecture (before OSL) to achieve a faster network with enhanced performance.

This article is organized as follows. Section 2 deals with the related work, whereas the proposed OSL-based deep acute leukemia detection model is presented in Section 3. A brief analysis of datasets is given in Section 4, whereas Section 5 deals with results and discussion. Finally, the conclusion is presented in Section 6.

2 Related Work

Currently, many researchers have developed several models for an efficient detection of acute leukemia. An efficient detection technique is developed by Das et al. [17] to differentiate between healthy and unhealthy lymphocytes affected by ALL. In this technique, gray-level run-length matrix (GLRLM) and Gray-level co-occurrence matrix (GLCM)-based texture, color, and shape features are extracted. Then, more important features are chosen using the Principal Component Analysis. Finally, SVM is used for ALL classification. Mishra et al. [18] have proposed a Random Forest-based ALL classification model using the ALLIDB1 dataset. Mishra et al. [32] have presented an SVM-based model for the detection of ALL using the ALLIDB1 dataset. Here, GLRLM is utilized to extract the features from the nucleus of the lymphocytic cell. In machine learning approaches, segmentation has a key role in overall performance enhancement. For this purpose, Das et al. [1] have developed a hybrid ellipse fitting (EF) based segmentation method for segmenting the blood cells to detect various hematological disorders. This approach results in more effective segmentation with a precise EF.

Mohammed and Abdulla [33] have recommended an effective system for detecting ALL. They have extracted important and necessary features like discrete cosine transform (DCT), shape, etc. Finally, classification is carried out using different techniques like KNN, NaÃ̃ve Bayes, or SVM. Shafique et al. [34] have introduced an SVM-based automated system for detecting ALL with the ALLIDB1 dataset. The method first segments the WBC, RBC, and platelets from the image and further differentiates the Lymphocytes from the WBC. Next, the color and shape features are extracted. An SVM-based computer-aided classification system has been proposed by Rawat et al. [35] for the detection of AML and ALL. This method first segments the nucleus from the myeloid and lymphoid cells. Further, the color and texture features are extracted from it.

An automated system has been presented by Agaian et al. [19] for the diagnosis of AML. It includes several steps: color correlation, segmentation of nucleus from the myeloid cells, feature extraction, and finally the classification. Color, shape, and local binary pattern (LBP)-based texture features are also extracted from the cell to get better classification results. Madhukar et al. [22] have applied SVM to classify AML successfully using the ASH dataset. Goutam and Sailaja [21] have recommended an effective and accurate system for the detection of AML using ASH dataset. The technique uses a K-means clustering algorithm for segmentation purposes, a Local directional path (LDP) for

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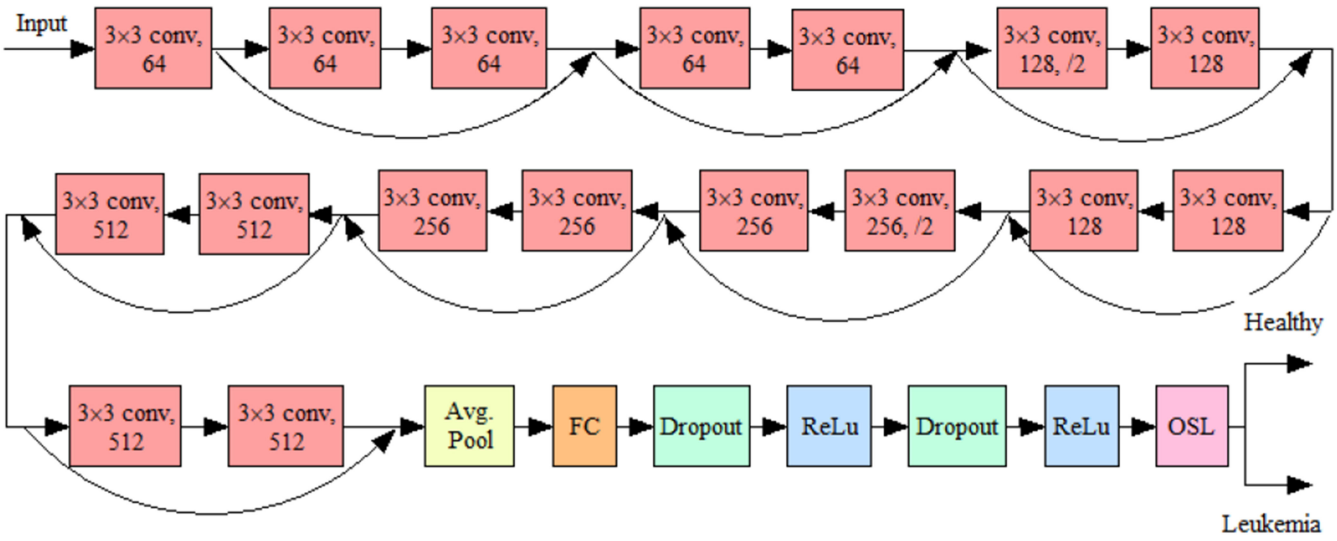


Fig. 1. Schematic of the proposed OSL-based acute leukemia detection model.

extracting the features, and finally SVM method for classification of the abnormality. Kumar and Udwadia [20] have presented an automatic AML detection system. This system first pre-processes and segments out the nucleus from the images using the K-means algorithm. In the next stage, the shape and texture features are extracted, and finally, the algorithm classifies the image into leukemic or non-leukemic classes using an SVM classifier.

However, to increase the efficiency of the system, new algorithms on deep learning have been proposed by researchers recently. In traditional deep learning-based systems, a large amount of training data is required for proper training to yield efficient feature extraction/ classification. Hence, transfer learning (TL) has grown as a preferred scheme for effective disease detection because of its exemplary performance on small datasets. Vogado et al. [13] have recommended an ALL identification model in which pre-trained TL models (AlexNet, CafeNet, or VGG-f) are applied to extract features. Then SVM is used to achieve effective classification. In 2021, Das et al. [15] have suggested a ShuffleNet-based ALL classification technique. It has achieved a 96.97% accuracy and 98.00% recall in the ALLIDB1 dataset, whereas a 96.67% accuracy and 96.46% recall in the ALLIDB2 dataset. Das et al. [14] have presented an effective modified ResNet50-based feature extraction model for detecting leukemia, where logistic regression, SVM, and random forest (RF) are used to yield efficient classification. In 2021, a new hybrid transfer learning (MobileNetV2-ResNet18)-based ALL identification system is developed by Das et al. [2]. This hybrid model has achieved a 97.18% accuracy and 95.90% recall in the ALLIDB2 dataset. A new CNN-based algorithm is now proposed that uses transfer learning and orthogonal softmax layer to improve the classification performance further, i.e., to achieve better classification results as compared to the algorithms already developed [2], [14], [15].

From the above discussion, we observe that conventional machine learning-based acute leukemia schemes require additional segmentation schemes to deliver promising performance. Recent advancements in deep learning, particularly transfer learning, make acute leukemia detection more efficient as it can provide promising performance even in small datasets, unlike conventional deep learning schemes. In addition, it does not need extra segmentation schemes, unlike conventional machine learning schemes. Among these available schemes discussed in this literature, the ALL detection scheme suggested in [33] and [2] depict the best performances in ALLIDB1 and ALLIDB2 datasets, respectively. Similarly, in the ASH dataset, the AML detection framework recommended by Agaian et al. [19] delivers the best performance among the schemes discussed in this

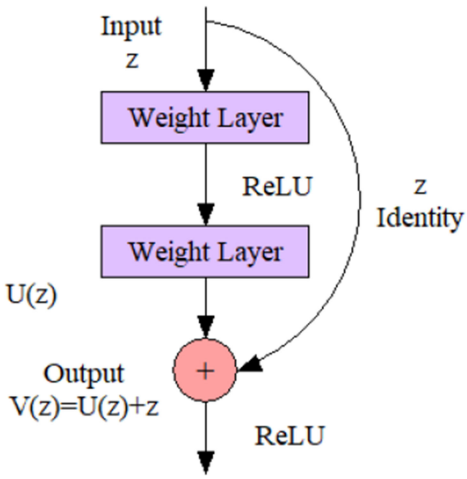


Fig. 2. Residual building block.

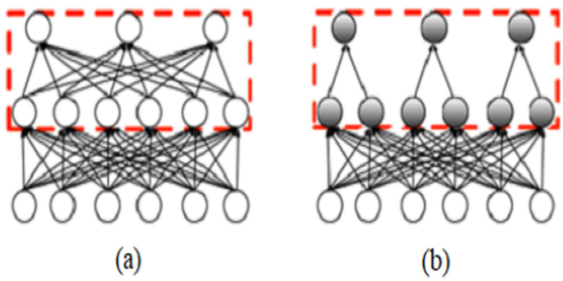


Fig. 3. Classification layer [11]: (a) A standard CNN model with FCCL; (b) A CNN model with OSL (as shown in dotted line).

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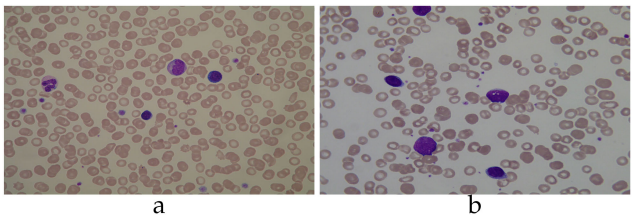


Fig. 4. ALLIDB1 dataset images: (a) Healthy and (b) ALL.

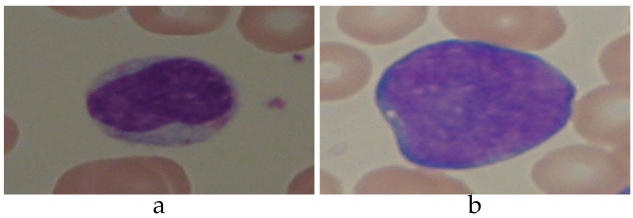


Fig. 5. ALLIDB2 dataset image: (a) Healthy and (b) ALL.

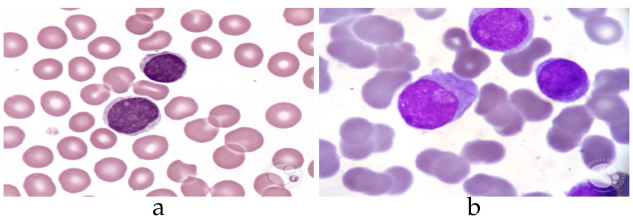


Fig. 6. ASH dataset images: (a) Non-AML and (b) AML.

literature. The success of transfer learning encourages us to develop a more efficient transfer learning and OSL-based acute leukemia detection system.

TABLE 1
Performance Comparison in the ALLIDB1 Dataset

Method	Accuracy (%)	Recall (%)	Specificity (%)
[34]	93.70	92.00	91.00
[18]	96.00	86.50	99.56
[17]	96.00	92.64	93.06
[32]	96.97	87.00	**100.00**
[33]	97.45	–	–
[15]	96.97	98.00	96.11
Proposed	**99.39**	**98.65**	**100.00**

TABLE 2
Performance Comparison in the ALLIDB2 Dataset

Method	Accuracy (%)	Recall (%)	Specificity (%)
[45]	95.67	–	–
[46]	96.01	–	–
[14]	96.15	95.65	96.55
[2]	97.18	95.90	**98.46**
[15]	96.67	96.46	96.90
Proposed	**98.21**	**97.95**	**98.46**

TABLE 3
Performance Comparison in the ASH Dataset

Method	Accuracy (%)	Recall (%)	Specificity (%)
[22]	93.50	83.00	91.00
[20]	95.00	–	–
[19]	97.45	–	–
Proposed	**97.50**	**96.67**	**98.33**

3 Proposed Method

This work presents a novel Orthogonal SoftMax Layer (OSL)-based lighter deep learning model for efficiently detecting acute leukemia. It consists of ResNet 18-based deep feature extraction followed by efficient OSL-based classification. The OSLNet approach recommended by Li et al. [11] exhibits very good small-sample classification performance, where VGG16 is employed for feature extraction, and OSLNet is applied for effective classification. As the OSLNet [11] is observed to be a very efficient classifier for small datasets [36], it becomes the source of inspiration for developing a still better classifier for detecting ALL and AML, for which available datasets are usually small.

Fig. 1 depicts the schematic of the proposed OSL-based acute leukemia detection model. In this model, feature extraction is done using computationally efficient ResNet18 instead of VGG16. OSL replaces a FCCL to boost the classification performance and makes the network faster. Hence, it integrates the benefits of both ResNet and OSL-based classification. Furthermore, extra dropout and ReLu layers are introduced in the proposed architecture (before OSL) to achieve a faster network with enhanced performance. With dropout, the accuracy will gradually increase, and loss will gradually decrease. It prevents a model from overfitting and makes the model faster. The proposed method mainly consists of two parts: -

1. ResNet18-based feature extraction
2. Orthogonal SoftMax Layer (OSL)-based Classification

3.1 ResNet18-Based Feature Extraction

In this proposed model, ResNet18 [37] is employed to extract more significant features. ResNet18 has eighteen weighted layers. ResNet [37] is developed by He et al. to solve the vanishing gradient issue in a deep CNN model. For this purpose, they have recommended skip connection or identity mapping, as shown in Fig. 2. It gives a substitute path to flow the gradient.

Here, the input of the first convolution (z) is directly given to the output of the second convolution. Hence, $(V(z))$ is expressed as

$$V(z) = U(z) + z. \tag{1}$$

The residual $U(z)$ is rewritten as

$$U(z) = V(z) - z. \tag{2}$$

In ResNet, residual $U(z)$ is learned. It is a small perturbation of the input (z). Thus, residual learning makes ResNet computationally efficient. On the other hand, identity mapping gives a substitute path to flow the gradient, resulting

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TABLE 4
ALL Classification 10-Fold Cross-Validation Performance Using ALLIDB1 Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
VGG16	78.67	82.27	83.50	81.33	0.8101	0.8242
VGG16-OSLNet	86.17	87.82	87.50	88.00	0.8683	0.8775
VGG19	79.00	81.36	81.50	81.33	0.8023	0.8142
VGG19-OSLNet	84.67	85.18	84.00	86.67	0.8433	0.8534
ResNet50	88.67	91.64	94.00	89.67	0.9126	0.9184
ResNet50-OSLNet	92.33	93.64	94.00	93.33	0.9316	0.9367
MobileNetV2	96.33	95.45	94.00	96.67	0.9515	0.9533
MobileNetV2-OSLNet	96.67	97.27	**98.00**	96.67	0.9733	0.9733
ResNet18	94.67	96.36	**98.00**	95.00	0.9630	0.9650
Proposed	**100.00**	**99.09**	**98.00**	**100.00**	**0.9899**	**0.9900**

TABLE 5
ALL Classification 10-Fold Cross-Validation Performance Using ALLIDB2 Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
VGG16	85.05	86.15	87.69	84.62	0.8635	0.8615
VGG16-OSLNet	87.69	89.62	92.31	86.92	0.8994	0.8962
VGG19	84.95	85.77	86.92	84.62	0.8592	0.8577
VGG19-OSLNet	91.86	89.62	86.92	92.31	0.8932	0.8962
ResNet50	89.67	90.77	92.31	89.23	0.9097	0.9077
ResNet50-OSLNet	93.01	92.31	91.54	93.08	0.9227	0.9231
MobileNetV2	95.38	94.23	93.08	95.38	0.9422	0.9423
MobileNetV2-OSLNet	98.46	96.15	93.85	**98.46**	0.9610	0.9615
ResNet18	94.78	95.00	95.38	94.62	0.9508	0.9500
Proposed	**98.57**	**98.08**	**97.69**	**98.46**	**0.9813**	**0.9808**

in more efficient training of deeper networks. Hence, it enhances the performance.

3.2 Orthogonal SoftMax Layer (OSL)-Based Classification

OSL is an efficient classification layer that replaces a FCCL to boost classification performance and makes the network faster, as shown in Fig. 3. For our understanding, we write a CNN model with OSL as OSLNet. In this layer, the weight vectors are kept orthogonal to each other.

The OSL is similar to the Dropout and DropConnect [11], [38] methods which also drops few connections in the model. Dropout randomly sets the chosen subset of neurons/nodes to zero in each layer [11], [25], whereas DropConnect removes a randomly selected subset of weight vectors within a network. Both Dropout and DropConnect decrease the co-adaptation within the neural network.

If an input vector (X) and an output vector (Y) are given by

$$X = [x_1, x_2, x_3, \ldots, x_D]^T \tag{3}$$

$$Y = [y_1, y_2, y_3, \ldots, y_n]^T, \tag{4}$$

then an FCCL may be expressed as

$$Y = a(WX), \tag{5}$$

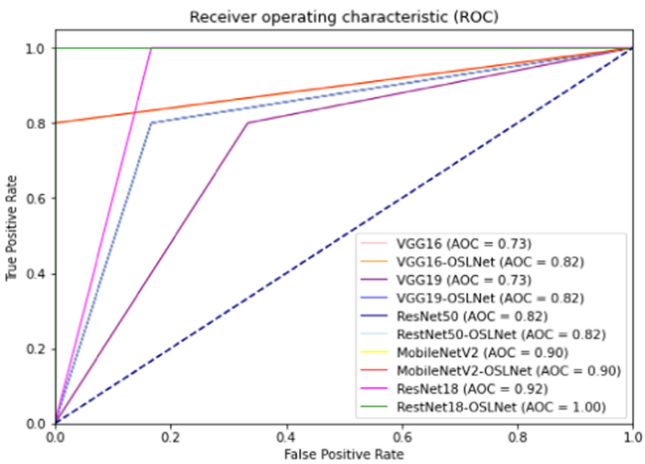


Fig. 7. ROC performance with ALLIDB1 dataset.

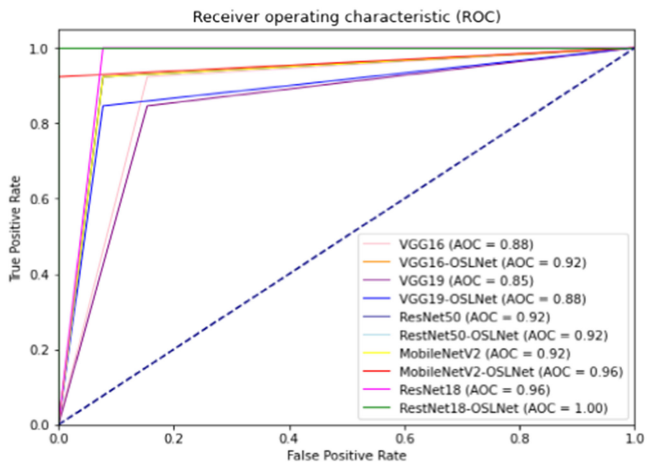


Fig. 8. ROC performance with ALLIDB2 dataset.

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TABLE 6
AML Classification 10-Fold Cross-Validation Performance Using ASH Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
VGG16	86.50	85.00	85.00	85.00	0.8574	0.8500
VGG16-OSLNet	91.50	87.50	85.00	90.00	0.8813	0.8750
VGG19	84.00	82.50	82.50	82.50	0.8324	0.8250
VGG19-OSLNet	85.00	85.00	87.50	82.50	0.8623	0.8500
ResNet50	91.00	90.00	90.00	90.00	0.9050	0.9000
ResNet50-OSLNet	93.50	93.75	95.00	92.50	0.9424	0.9375
MobileNetV2	94.00	93.75	95.00	92.50	0.9450	0.9375
MobileNetV2-OSLNet	96.00	96.25	**97.50**	95.00	0.9674	0.9625
ResNet18	96.00	93.75	92.50	95.00	0.9422	0.9375
Proposed	**98.00**	**97.50**	**97.50**	**97.50**	**0.9775**	**0.9750**

where, a symbolizes the activation function, and W denotes an $n \times D$ weight matrix.

Then, an FCCL with Dropout may be expressed as

$$Y = m * a(WX), \tag{6}$$

where, $*$ symbolizes the element-wise multiplication and m denotes a n-dimension binary vector. The ith element of m is represented as

$$m_i \sim Bernoulli(1-p), i \in \{1, 2, \ldots n\}, \tag{7}$$

where, p represents the probability of dropping a neuron.

Then, the FCCL with Dropconnect is given by

$$Y = a((M * W)X), \tag{8}$$

Each entry of binary matrix (M) is represented as

$$M_{ji} \sim Bernoulli(1-q), j \in \{1, 2, \ldots n\}, \tag{9}$$

where, q denotes the a probability of dropping a connection.

The objective of OSL is to retain large angles between weights by dropping few connections in the FCCL and making the weight vectors orthogonal to each other [11]. The OSL is expressed as

$$Y = softmax((M * W)X), \tag{10}$$

The matrix M plays the role of masking and hence is termed a mask matrix. To make the weight vectors orthogonal to each other, M should be a binary matrix. Here, $M_{j,i}$ is one if j=i, otherwise $M_{j,i}$ is zero. Thus, an orthogonal weight matrix ($M * W$) is obtained. In OSL, the orthogonality results in improvement of computational efficiency by minimizing co-adaption among parameters.

Some salient features of OSL are mentioned here.

1. OSL drops few weight vectors connections and retains the same throughout the training and testing. In a further round, it fixes the architecture in both training and testing stages. However, DropConnect randomly drops some connections with a specified probability only in the training stage and not in the testing stage.
2. OSL can be efficiently used as a classification layer, unlike Dropconnect. OSL optimizes the number of connections by eliminating a few connections, which makes it computationally efficient. On the other hand, DropConnect is not recommended as a classification layer due to the fact that the whole or part of the network cannot be trained if all the output neuron connections are removed.
3. In OSL, every neuron is assigned to a particular class before the training is started. Hence, the training of the model becomes easier. Furthermore, the solution space of a model with OSL is a subset of that of the same model with an FCCL. It signifies that OSL makes the model computationally efficient.

4 Datasets

Experiments are carried out by employing standard ALLIDB1 [39] and ALLIDB2 [39] datasets for detecting ALL and an ASH [40] dataset for detecting AML. ALLIB1 has 59 healthy and 49 sick (ALL) images (multiple leukocytes per image). ALLIB2 has 130 healthy and 130 sick (ALL) images (single leukocyte per image). ALLIDB1 and ALLIDB2 are publicly available standard datasets, which are available at https://homes.di.unimi.it/scotti/all/.

The standard ASH [40] dataset is available at http://imagebank.hematology.org. For the effective detection of AML, 40 AML and 40 Non-AML images are used. Figs. 4, 5, and 6 display some images of ALLIDB1, ALLIDB2, and ASH datasets, respectively. In addition, the standard C_NMC_2019 dataset [41] is also employed to make a comparative leukemia (ALL) detection performance analysis better. It is publicly available at https://doi.org/10.7937/tcia.2019.dc64i46r. It contains 15135 images from 118 patients. It consists of two classes of images: leukemia and healthy images [41], [42], [43], [44]. Here, we consider 150 healthy and 150 leukemia images to validate the performance of proposed schemes even in the presence of small datasets.

5 Result and Discussion

This section delivers a comparative performance analysis for the efficient detection of acute leukemia. For this purpose, performance analysis is performed using 10-Fold cross-validation. The average testing performance is illustrated for comparative performance analysis. Nevertheless, the fifth Fold's receiver operating characteristic (ROC) performance is presented to display the suggested ResNet-OSL

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TABLE 7
ALL Classification 10-Fold Cross-Validation Performance Using C_NMC_2019 Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
Vgg16	78.08	77.33	76.00	78.67	0.7703	0.7733
Vgg16-OSLNet	80.00	79.00	7900	80.67	0.7864	0.7900
Vgg19	78.91	78.33	77.33	79.33	0.7811	0.7833
Vgg19-OSLNet	79.87	79.67	79.33	80.00	0.7960	0.7967
Resnet50	85.81	85.33	84.67	86.00	0.8524	0.8533
Resnet50-OSLNet	87.16	86.67	86.00	87.33	0.8658	0.8667
MobileNetV2	87.07	86.33	85.33	87.33	0.8619	0.8633
MobileNetV2-OSLNet	89.66	88.33	86.67	90.00	0.8814	0.8833
Resnet18	88.36	87.33	86.00	88.67	0.8716	0.8733
Resnet18-OSLNet	**91.03**	**89.67**	**88.00**	**91.33**	**0.8949**	**0.8967**

model's superiority. Moreover, a comparative performance analysis is also performed using mean performances of five consecutive experiments with 70% data for training and 30% data for testing. The experimental work is performed on Google Colab, whose specifications are as follows.

a) Intel(R) Xeon(R) CPU @ 2.20 GHz
b) 1xTesla K80 processor with 2496 CUDA cores
c) 12 GB RAM
d) 33 GB Disk Space.

Tables 1, 2, and 3 display a relative performance comparison with several existing methods in ALLIDB1, ALLIDB2, and ASH datasets, respectively. The proposed ResNet18-OSL framework achieves 99.39% accuracy in the ALLIDB1 dataset, which is 1.94% higher than the best-performing existing scheme, [33]. Similarly, in the ALLIDB2 dataset, the proposed ResNet18-OSL framework depicts the best performance with 98.21% accuracy, whereas the hybrid transfer learning network suggested in [2] delivers 97.18% accuracy, which is the best among existing schemes. In ASH datasets, the proposed ResNet18-OSL framework also outperforms other existing works with 97.50% accuracy. Overall, the suggested ResNet18-OSL framework delivers the best accuracy, recall, and specificity performance in all these datasets, which makes it a preferred candidate for acute leukemia detection.

Tables 4 and 5 represent the average ALL classification performance (10-Fold cross-validation performance) in ALLIDB1 and ALLIDB2 datasets, respectively. The corresponding ROC performances are illustrated in Figs. 7 and 8, respectively. From both the tables, we observe the proposed ResNet-OSL model achieves excellent performance due to the integrated advantages of ResNet and OSL. The proposed framework outperforms others as the orthogonal softmax layer (OSL) replaces FCCL, which not only makes the system faster but also improves the classification performance (since few connections are removed to make weight vectors orthogonal to each other). Thus, it minimizes the co-adaption among parameters and results in reduced generalization error compared to an FCCL. Moreover, it optimizes the correct predictions (true positives and true negatives) and minimizes the false predictions (false positives and false negatives) due to the improvement of feature discrimination capability because of inter-class angle maximization in OSL. Thus, the proposed ResNet18-OSL framework achieves the best performance in terms of accuracy, precision, specificity, sensitivity, AUC, and F1 Score, which can also be observed from these tables. In ALLIDB1 datasets, it delivers 99.09% accuracy, whereas the second-best performing model, i.e., MobileNetV2-OSL, achieves 97.27% accuracy. Similarly, in ALLIDB2 datasets, the proposed ResNet18-OSL outperforms others with 98.08% accuracy, whereas MobileNetV2-OSL stands second with 96.15% accuracy. Moreover, the employment of OSL in place of FCCL improves the accuracy performance by around 2.73% and 3.08% in ALLIDB1 and ALLIDB2 datasets, respectively. Figs. 7 and 8 also indicate the excellence of the ResNet-OSL model. Tables 8 and 9 present the ALL detection performance (10-Fold cross-validation performance) in terms of TN, FP, FN, and TP with ALLIDB1 and ALLIDB2 datasets, respectvely. Both tables indicate the

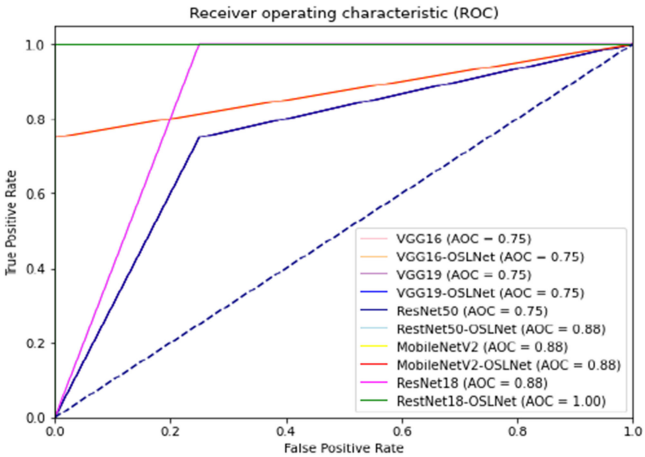


Fig. 9. ROC performance with ASH dataset.

TABLE 8
TN, FP, FN, and TP Performances in ALLIDB1 Dataset

Method	TN	FP	FN	TP
VGG16	48	11	8	41
VGG16-OSLNet	52	7	6	43
VGG19	48	11	9	40
VGG19-OSLNet	51	8	8	41
ResNet50	53	6	3	46
ResNet50-OSLNet	55	4	3	46
MobileNetV2	57	2	3	46
MobileNetV2-OSLNet	57	2	1	48
ResNet18	56	3	1	48
Proposed	59	0	1	48

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TABLE 9
TN, FP, FN, and TP Performances in ALLIDB2 Dataset

Method	TN	FP	FN	TP
VGG16	110	20	16	114
VGG16-OSLNet	113	17	10	120
VGG19	110	20	17	113
VGG19-OSLNet	120	10	17	113
ResNet50	116	14	10	120
ResNet50-OSLNet	121	9	11	119
MobileNetV2	124	6	9	121
MobileNetV2-OSLNet	128	2	8	122
ResNet18	123	7	6	124
Proposed	128	2	3	127

TABLE 10
TN, FP, FN, and TP Performances in ASH Dataset

Method	TN	FP	FN	TP
VGG16	34	6	6	34
VGG16-OSLNet	36	4	6	34
VGG19	33	7	7	33
VGG19-OSLNet	33	7	5	35
ResNet50	36	4	4	36
ResNet50-OSLNet	37	3	2	38
MobileNetV2	37	3	2	38
MobileNetV2-OSLNet	38	2	1	39
ResNet18	38	2	3	37
Proposed	39	1	1	39

superiority of the proposed schemes with minimum false detections.

From Table 6, we observe the ResNet-OSL model achieves superior AML classification performance (10-Fold cross-validation performance) with 97.50% accuracy, 98.00% precision, and 97.50% specificity. On the other hand, MobileNetV2-OSL stands second with 96.25% accuracy, 96.00% precision, and 95.00% specificity. Moreover, the proposed ResNet18-OSL framework achieves a 3.75% improvement in accuracy over the ResNet18 model due to the benefits of OSL-based classification (minimization of the co-adaption among parameters, minimization of the generalization error, and improvement of feature discrimination capability). Fig. 9 illustrates ResNet-OSL model yields excellent AML classification performance in terms of ROC. Table 10 represents the AML detection performance (10-Fold cross-validation performance) in terms of TN, FP, FN, and TP with ASH dataset. It indicates the proposed schemes exhibits the best performance among all with minimum false detections.

Moreover, the 10-Fold cross-validation of the proposed scheme and the comparing schemes are validated with a more challenging standard C_NMC_2019 dataset [41]. Table 7 presents the comparative performance analysis of the proposed scheme in this dataset. The corresponding ROC performance is illustrated in Fig. 10. From this table, we observe that the proposed ResNet18-OSLNet-based leukemia detection scheme outperforms others with 89.67% accuracy, 91.03% precision, 88.00% sensitivity, 91.33% specificity, 0.8949 F1 score, and 0.8967 AUC. For better analysis, Table 11 presents the performance in terms of TN, FP, FN, and TP values. From the table, it is found that the proposed scheme exhibits the best performance among all with the least number of false detections(FP and FN). It demonstrates the best results due to the integrated merits of ResNet18 and OSL.

In addition, a comparative performance analysis is also performed using mean performances of five consecutive experiments with 70-30 data distribution (70% data for training and 30% data for testing).

Tables 12, 13, 14, and 15 present the performances (with 70-30 data distribution) in ALLIDB1, ALLIDB2, ASH, and $C-NMC-2019$ datasets, respectively. From these tables, it is observed that the proposed ResNet-OSL framework outperforms others with the best quantitative measures in all four datasets. The MobileNetV2-OSL framework demonstrates the second-best classification performance in all four datasets. The proposed framework delivers 1.84%, 0.77%, 0.83%, and 1.12% improvement over the second-best performing MobileNetV2-OSL scheme in ALLIDB1, ALLIDB2, ASH, and $C-NMC-2019$ datasets, respectively by integrating the benefits of ResNet18 with OSL. Moreover, the proposed framework yields the best correct predictions with the least false predictions in all four datasets, as shown in Tables 16, 17, 18, and 19.

Table 20 presents the average testing time (in seconds) per fold in 10-Fold cross-validation, whereas Table 21

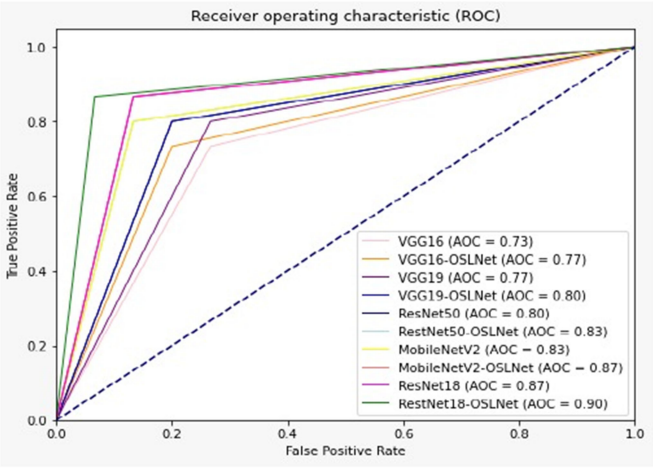


Fig. 10. ROC performance with $C-NMC-2019$ dataset.

TABLE 11
TN, FP, FN, and TP Performances (10-Fold Cross-Validation) in C_NMC_2019 Dataset

Method	TN	FP	FN	TP
Vgg16	118	32	36	114
Vgg16-OSLNet	121	29	34	116
Vgg19	119	31	34	116
Vgg19-OSLNet	120	30	31	119
Resnet50	129	21	23	127
Resnet50-OSLNet	131	19	21	129
MobileNetV2	131	19	22	128
MobileNetV2-OSLNet	135	15	20	130
Resnet18	133	17	21	129
Resnet18-OSLNet	**137**	**13**	**18**	**132**

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TABLE 12
ALL Classification Performance (With 70-30 Data Distribution) Using ALLIDB1 Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
VGG16	84.42	87.12	87.84	86.52	86.10	0.8718
VGG16-OSLNet	89.47	91.41	91.89	91.01	90.66	0.9145
VGG19	83.12	85.89	86.49	85.39	84.77	0.8594
VGG19-OSLNet	88.16	90.18	90.54	89.89	89.33	0.9021
ResNet50	93.33	94.48	94.59	94.38	93.96	0.9449
ResNet50-OSLNet	95.95	96.32	95.95	96.63	95.95	0.9629
MobileNetV2	94.67	95.71	95.95	95.51	95.31	0.9573
MobileNetV2-OSLNet	97.30	97.55	97.30	97.75	97.30	0.9753
ResNet18	97.26	96.93	95.95	97.75	96.60	0.9685
Proposed	**100.00**	**99.39**	**98.65**	**100.00**	**99.32**	**0.9932**

TABLE 13
ALL Classification Performance (With 70-30 Data Distribution) Using ALLIDB2 Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
VGG16	89.39	90.00	90.77	89.23	90.07	0.9000
VGG16-OSLNet	91.88	92.31	92.82	91.79	92.35	0.9231
VGG19	88.44	89.23	90.26	88.21	89.34	0.8923
VGG19-OSLNet	91.37	91.79	92.31	91.28	91.84	0.9179
ResNet50	92.42	93.08	93.85	92.31	93.13	0.9308
ResNet50-OSLNet	94.42	94.87	95.38	94.36	94.90	0.9487
MobileNetV2	96.34	95.38	94.36	96.41	95.34	0.9538
MobileNetV2-OSLNet	97.93	97.44	96.92	97.95	97.42	0.9744
ResNet18	96.89	96.41	95.90	96.92	96.39	0.9641
Proposed	**98.45**	**98.21**	**97.95**	**98.46**	**98.20**	**0.9821**

TABLE 14
AML Classification Performance (With 70-30 Data Distribution) Using ASH Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
VGG16	88.14	87.50	86.67	88.33	87.40	0.8750
VGG16-OSLNet	89.83	89.17	88.33	90.00	89.07	0.8917
VGG19	87.93	86.67	85.00	88.33	86.44	0.8667
VGG19-OSLNet	89.66	88.33	86.67	90.00	88.14	0.8833
ResNet50	93.10	91.67	90.00	93.33	91.52	0.9167
ResNet50-OSLNet	95.00	95.00	95.00	95.00	95.00	0.9500
MobileNetV2	93.44	94.17	95.00	93.33	94.21	0.9417
MobileNetV2-OSLNet	96.67	96.67	96.67	96.67	96.67	0.9667
ResNet18	96.55	95.00	93.33	96.67	94.91	0.9500
Proposed	**98.31**	**97.50**	**96.67**	**98.33**	**97.48**	**0.9750**

TABLE 15
ALL Classification Performance (With 70-30 Data Distribution) Using $C-NMC-2019$ Dataset

Method	Precision (%)	Accuracy (%)	Recall (%)	Specificity (%)	F1 Score	AUC
VGG16	84.23	83.78	83.11	84.44	83.67	0.8378
VGG16-OSLNet	85.97	85.33	84.44	86.22	85.20	0.8533
VGG19	83.86	83.56	83.11	84.00	83.48	0.8356
VGG19-OSLNet	86.30	85.33	84.00	86.67	85.13	0.8533
ResNet50	88.64	87.78	86.67	88.89	87.64	0.8778
ResNet50-OSLNet	89.59	88.89	88.00	89.78	88.79	0.8889
MobileNetV2	89.09	88.22	87.11	89.33	88.09	0.8822
MobileNetV2-OSLNet	91.36	90.44	89.33	91.56	90.33	0.9044
ResNet18	90.05	89.33	88.44	90.22	89.24	0.8933
Proposed	**92.31**	**91.56**	**90.67**	**92.44**	**91.48**	**0.9156**

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TABLE 16
TN, FP, FN, and TP Performances (With 70-30 Data Distribution) in ALLIDB1 Dataset

Method	TN	FP	FN	TP
VGG16	77	12	9	65
VGG16-OSLNet	81	8	6	68
VGG19	76	13	10	64
VGG19-OSLNet	80	9	7	67
ResNet50	84	5	4	70
ResNet50-OSLNet	86	3	3	71
MobileNetV2	85	4	3	71
MobileNetV2-OSLNet	87	2	2	72
ResNet18	87	2	3	71
Proposed	89	0	1	73

TABLE 17
TN, FP, FN, and TP Performances (With 70-30 Data Distribution) in ALLIDB2 Dataset

Method	TN	FP	FN	TP
VGG16	174	21	18	177
VGG16-OSLNet	179	16	14	181
VGG19	172	23	19	176
VGG19-OSLNet	178	17	15	180
ResNet50	180	15	12	183
ResNet50-OSLNet	184	11	9	186
MobileNetV2	188	7	11	184
MobileNetV2-OSLNet	191	4	6	189
ResNet18	189	6	8	187
Proposed	192	3	4	191

TABLE 18
TN, FP, FN, and TP Performances (With 70-30 Data Distribution) in ASH Dataset

Method	TN	FP	FN	TP
VGG16	53	7	8	52
VGG16-OSLNet	54	6	7	53
VGG19	53	7	9	51
VGG19-OSLNet	54	6	8	52
ResNet50	56	4	6	54
ResNet50-OSLNet	57	3	3	57
MobileNetV2	56	4	3	57
MobileNetV2-OSLNet	58	2	2	58
ResNet18	58	2	4	56
Proposed	59	1	2	58

depicts the average testing time (with 70-30 data distribution) in seconds. Both the tables indicate the proposed ResNet18-OSL is faster than others, whereas MobileNetV2-OSL ranks second in all four datasets. It is also observed that the ResNet-OSL framework proposed here has a very high throughput rate (highest among all compared here). Moreover, it yields the best performance results in terms of precision, accuracy, recall, specificity, F1 score, and AUC among all the methods mentioned in these tables. Nevertheless, the proposed ResNet-OSL model not only achieves excellent performance but also computationally faster than others.

TABLE 19
TN, FP, FN, and TP Performances (With 70-30 Data Distribution) in $C-NMC-2019$ Dataset

Method	TN	FP	FN	TP
VGG16	190	35	38	187
VGG16-OSLNet	194	31	35	190
VGG19	189	36	38	187
VGG19-OSLNet	195	30	36	189
ResNet50	200	25	30	195
ResNet50-OSLNet	202	23	27	198
MobileNetV2	201	24	29	196
MobileNetV2-OSLNet	206	19	24	201
ResNet18	203	22	26	199
Proposed	208	17	21	204

TABLE 20
Testing Time Per Fold (In Seconds) Per Fold in 10-Fold Cross-Validation

Method	ALL-IDB1	ALL-IDB2	ASH	C_NMC_2019 IDB1
VGG16	10.88	25.12	9.01	24.04
VGG16-OSLNet	10.70	24.94	8.87	23.72
VGG19	10.97	25.25	9.04	24.31
VGG19-OSLNet	10.77	25.20	8.92	24.15
ResNet50	7.98	21.97	6.11	20.89
ResNet50-OSLNet	7.88	21.82	6.03	20.71
MobileNetV2	7.90	21.86	6.05	20.74
MobileNetV2-OSLNet	7.80	21.71	5.98	20.58
ResNet18	7.83	21.77	6.02	20.64
Proposed	**7.74**	**21.63**	**5.93**	**20.49**

TABLE 21
Average Testing Time (With 70-30 Data Distribution) in Second

Method	ALL-IDB1	ALL-IDB2	ASH	C_NMC_2019 IDB1
VGG16	32.17	73.92	26.83	71.48
VGG16-OSLNet	31.74	73.41	26.41	70.61
VGG19	32.48	74.39	26.94	72.25
VGG19-OSLNet	31.89	73.92	26.58	71.82
ResNet50	23.65	65.73	18.19	62.17
ResNet50-OSLNet	23.36	65.31	18.08	61.73
MobileNetV2	23.49	65.46	18.13	61.80
MobileNetV2-OSLNet	23.17	65.04	17.94	61.46
ResNet18	23.24	65.17	18.04	61.51
Proposed	**22.89**	**64.91**	**17.78**	**61.23**

6 Conclusion

In this paper, a novel OSL-based acute leukemia classification model is proposed, where OSL is integrated with the ResNet18 to improve the classification performance by making the weight vectors orthogonal to each other. Hence, it retains the benefits of both ResNet (residual learning and identity mapping) and OSL (improvement of feature discrimination capability and computational efficiency). The simulation results also indicate that the proposed ResNet-OSL model achieves excellent performance and is faster

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than others. Though the proposed ResNet18-OSL outperforms other existing methods discussed here, there is a scope for further performance improvement. The performance can be further improved by extracting diverse features, i.e., features at different resolutions and sampling rates. Integrated benefits of sparse learning, transfer learning, and OSL may also enhance classification performance.

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